



Research Proposal for a Master's Thesis

Studies:	Data Science
Student:	Emirhan Kurtulus
Address:	Absberggasse 25/6/3
Tel.Nr.:	+43 699 1919 1954
Email:	e12243493@student.tuwien.ac.at
Matriculation Number:	12243493
Thesis Advisor:	Univ.-Prof. Dr.-Ing. habil. Fazel Ansari
Content Support:	Dipl.-Ing. Andreas Steiner, B.Sc.
Planned Date of Exposé Submission:	30 th July 2026
Planned Time Period:	6 months

Preliminary Title: Structured Subsystem-Aware Feature Representations for Predictive Maintenance in Metro Systems

Confirmation of conclusion of the support agreement

Exposé confirmed on:

(Advisor Signature)

(Student Signature)

1 Title

Structured Subsystem-Aware Feature Representations for Predictive Maintenance in Metro Systems

2 Initial Situation: Motivation and Problem Statement

Introduction: The rapid digital transformation of industrial systems within the context of Industry 4.0 has enabled the continuous collection of high-frequency sensor data from complex technical infrastructures. This development has facilitated the shift from traditional maintenance strategies toward predictive maintenance, which uses condition monitoring data to detect changes in system behavior and anticipate failures before they occur. Predictive maintenance aims to improve system reliability, reduce operational costs, and minimize unplanned downtime by leveraging data-driven approaches.

In railway and metro systems, predictive maintenance is particularly important due to the safety-critical nature of operations and the high impact of service disruptions. Service-affecting failures (SFAs) directly influence system availability, passenger safety, and operational efficiency. However, implementing reliable predictive maintenance in real-world settings remains challenging due to rare failure events, temporal misalignment, contextual uncertainty, and high false-alarm rates [1, 2].

To address these challenges, machine learning and deep learning methods have been widely applied to multivariate time-series data generated by IoT sensors, which capture operational states such as pressure, temperature, vibration, and rotation [3]. Despite their potential, existing approaches often struggle to generalize under real-world conditions, particularly when applied to high-dimensional datasets.

A key limitation of current predictive maintenance approaches is that sensor variables are commonly processed as one unified high-dimensional input space without explicitly considering subsystem-level functional relationships. In complex metro systems, however, failures often emerge within specific operational subsystems such as pneumatic braking systems, air production units, or leveling components. Ignoring these functional structures may reduce signal coherence, amplify irrelevant noise, and limit both predictive robustness and interpretability. Recent systematic reviews on predictive maintenance emphasize that although machine learning models are increasingly applied to high-

dimensional industrial sensor data, the organization and representation of input features remain comparatively underexplored in real-world deployment settings [4]. To support predictive maintenance research under realistic operational conditions, the MetroAT dataset was introduced as a large-scale event-aware benchmark dataset containing subsystem-level sensor measurements and aligned failure records from metro operations [5]. Initial evaluations on MetroAT indicate that both supervised and unsupervised approaches face robustness and false-alarm challenges under realistic temporal evaluation settings [1].

Failure-related anomaly windows provide an opportunity to identify recurring operational patterns through clustering and other unsupervised learning techniques. The identified patterns can subsequently be verified by domain experts and used to construct subsystem-aware feature representations.

Given these challenges, there remains a lack of systematic methodologies for constructing subsystem-aware feature representations from recurring anomaly patterns in high-dimensional sensor data, and it is unclear how such structured representations influence predictive robustness and diagnostic reliability. Therefore, this thesis investigates failure prediction in the MetroAT dataset [5], with the aim of analyzing how anomaly-informed subsystem-aware feature organization affects model performance and false-alarm behavior in real-world metro predictive maintenance.

Background Research: With increasing maintenance costs, predictive maintenance (PdM) has become a central research area within data-driven reliability engineering and industrial artificial intelligence [6]. In industrial and transportation systems, machine learning approaches are widely applied to multivariate time-series data to detect anomalies and forecast failures before they occur [6, 7]. Additionally, deep learning methods are widely used for modeling multivariate time-series data in predictive maintenance. In particular, recurrent neural networks (RNNs), such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are commonly applied because they can capture temporal dependencies in sequential data [8, 9]. Beyond predictive maintenance applications, deep learning has emerged as one of the most successful approaches for time-series classification, Fawaz et al. provide a comprehensive empirical evaluation of deep learning architec-

tures for time-series classification and demonstrate that convolutional and residual neural network architectures can automatically learn discriminative temporal representations from raw sequential data while achieving state-of-the-art performance across a wide range of benchmark datasets [10]. These findings suggest that deep learning models are suitable not only for failure prediction but also for learning subsystem-related patterns from anomaly-centered sensor windows.

Besides recurrent models, convolution-based approaches have also been explored. Hybrid models, such as CNN-LSTM, combine local feature extraction with temporal modeling and have shown improved performance in some applications [3]. More recently, Temporal Convolutional Networks (TCN) have been proposed as an alternative approach. TCN models use causal and dilated convolutions to capture long-range dependencies and can avoid some limitations of recurrent models, such as training instability [11].

A systematic literature review by Carvalho et al.[4] highlights that Random Forest, Support Vector Machine, recurrent neural networks, and deep learning approaches are among the most frequently applied machine learning methods in predictive maintenance applications [4]. The review further emphasizes that predictive maintenance systems increasingly rely on high-dimensional multivariate sensor data generated under Industry 4.0 environments, while real-world deployment remains challenging due to data heterogeneity, noise, and application-specific variability. Importantly, the review notes that most studies primarily focus on algorithmic performance, whereas comparatively limited attention is given to how input data representations and feature organization influence predictive maintenance performance and robustness[1].

This suggests that not only the model architecture, but also the structural organization of sensor data may influence predictive maintenance performance. In particular, subsystem-aware structured feature representations may improve robustness, reduce cross-subsystem noise, and enhance interpretability in safety-critical maintenance systems. Both recurrent and convolutional sequence modeling approaches are considered suitable for handling multivariate time-series data under noisy real-world conditions; however, their sensitivity to subsystem-aware versus flat high-dimensional

representations remains insufficiently explored [4, 12].

In the railway and metro domain, predictive maintenance research focuses on safety-critical subsystems such as braking, traction, and pneumatic systems [13, 14]. Machine learning-based monitoring systems and digital twin approaches aim to improve availability and reduce service-affecting failures (SFAs) [15]. However, practical deployment remains challenging due to rare failure events, strong class imbalance, heterogeneous operating conditions, and limited high-quality labeled data [1, 2]. Many studies rely on proxy labels or simplified datasets, which limits generalization to real operational environments [16].

While existing research investigates temporal modeling strategies, supervised versus unsupervised paradigms, and architectural comparison, comparatively limited attention has been given to subsystem-aware feature representations and the structural organization of sensor inputs [1]. Most predictive maintenance approaches process sensor variables within flat high-dimensional input spaces without explicitly leveraging subsystem-level functional dependencies or operational structure. Consequently, it remains unclear whether subsystem-aware structured representations can improve robustness, reduce false-alarm behavior, and enhance generalization performance under realistic metro operational conditions.

Purpose Identification: In this research, I aim to develop a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows through clustering, pattern analysis, and expert verification. The resulting subsystem-aware feature representations are subsequently evaluated with respect to predictive maintenance performance, false-alarm behavior, and generalization [3, 6, 7].

In addition, the study analyzes how different temporal modeling approaches respond to structured subsystem-level versus full-feature representations in failure prediction tasks. The models are applied to the real-world MetroAT dataset, a benchmark dataset for data-driven maintenance in metro operations [5]. Prior evaluations on MetroAT highlight that machine learning models face robustness and false-alarm challenges in practical deployment scenarios [1].

The research is structured into two consecutive phases. In the first phase, recurring anomaly patterns are identified from MetroAT anomaly windows through clustering and pattern analysis and subsequently used to construct subsystem-aware feature representations with expert verification.

In the second phase, the resulting subsystem-aware feature representations are evaluated against conventional full-feature representations under identical experimental conditions to assess their influence on predictive maintenance performance, false-alarm behavior, and generalization.

This research is therefore:

- Problem oriented, due to limited deployment reliability in current models [1],
- Diagnostic in nature, focusing on the relationship between anomaly patterns and subsystem-level feature categories,
- Comparative, analyzing subsystem-aware versus conventional full-feature representations under controlled experimental conditions,
- Empirical, based on evaluation on a real-world benchmark dataset [5, 16]

For this thesis, the goal is not to develop a completely new deep learning architecture. Instead, the focus is on developing a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows, and evaluating how these representations influence predictive maintenance performance in safety-critical metro operations.

Identification of Key Issues:

The research is based on the following key issues:

- **P1: Limited reliability of current prediction models.** Previous studies on the MetroAT dataset, particularly the work by Steiner et al.[1], show that both classical machine learning methods (e.g., Random Forest) and deep learning approaches (e.g., LSTM-based models and autoencoders) suffer from high false-alarm rates and limited generalization performance on unseen data, especially under realistic temporal evaluation settings.
- **P2: Lack of methodologies for constructing subsystem-aware feature representations from recurring anomaly patterns** Although failure-related anomalies contain valuable information about degradation processes, existing predictive maintenance approaches rarely use these patterns to systematically construct subsystem-aware feature representations. Instead, sensor variables are typically processed as a single high-dimensional feature space without explicitly leveraging recurring anomaly patterns to organize features according to subsystem behaviour.

- **P3: Lack of structured and controlled evaluation of feature representations.** Most existing approaches use all available sensor variables as a single high-dimensional input space and do not systematically investigate how the construction of subsystem-aware feature representations based on functional subsystem relationships and domain knowledge influences prediction performance and robustness under controlled experimental conditions [5].

3 Research Question and Aim of the Thesis

Research Questions:

- **RQ1:** To what extent can subsystem-aware feature representations constructed from recurring anomaly patterns extracted from MetroAT anomaly windows produce subsystem structures that are verified by domain experts?
- **RQ2:** To what extent do subsystem-aware feature representations improve predictive maintenance performance compared with conventional full-feature representations under identical experimental conditions?

Research Objectives:

- **O1:** Analyze anomaly-centered sensor windows derived from the MetroAT anomaly detection framework and identify recurring anomaly patterns using clustering and pattern analysis techniques.
- **O2:** Develop a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns through clustering, pattern analysis, and expert verification.
- **O3:** Construct subsystem-specific datasets from the expert-verified subsystem-aware feature representations.
- **O4:** Evaluate the influence of subsystem-aware feature representations on predictive maintenance performance, robustness, and generalization using different temporal modeling approaches.

4 Methodology and Methodical Approach

1. Methods for implementing the research

This thesis follows an empirical and experimental research design. The study combines a structured literature review with a controlled dataset-based evaluation of temporal models for predictive maintenance.

1) Targeted Literature Review

A structured literature review will be conducted to identify relevant research on:

- Temporal modeling approaches for multivariate time-series failure prediction,
- Deployment challenges and strategies to improve robustness in real-world datasets (e.g., structured subsystem-level representations, subsystem-aware representations, anomaly detection).

The review process will include the following steps:

- **Search Strategy Development:** Conduct searches in IEEE Xplore, ACM Digital Library, SpringerLink, and Scopus using predefined keyword combinations related to predictive maintenance, multivariate time-series analysis, representation learning, subsystem-aware feature representations, anomaly detection, and real-world deployment challenges in railway and industrial systems.

The literature review particularly focuses on studies investigating:

- structured and domain-informed feature representations,
- temporal modeling approaches for multivariate sensor data,
- representation learning and feature organization strategies,
- anomaly-informed predictive maintenance methods,
- robustness, generalization, and false-alarm behavior in safety-critical environments,
- data-driven approaches for high-dimensional industrial sensor data analysis.

- **Screen and Select Relevant Literature:** Screen titles and abstracts based on relevance to safety-critical systems and time-series modeling. The selection process will follow a structured review approach (e.g., PRISMA-inspired filtering), prioritizing peer-reviewed journal and conference papers with strong methodological transparency.
- **Specific Data Extraction:** Extract information on:
 - Model architecture and structure,
 - Temporal modeling strategy,
 - Handling of class imbalance,
 - Evaluation metrics and reported performance,
 - Dataset characteristics,
 - Reported robustness and deployment considerations.
- **Reporting Format:** Organize selected studies in structured tabular form to compare modeling paradigms, strengths, and limitations. The synthesis will provide the basis for selecting suitable temporal modeling approaches for the experimental evaluation.

2) Experimental Dataset-Based Study

The core research method is a controlled experimental evaluation on the MetroAT dataset. The study develops and evaluates a methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows under real-world operational conditions.

The experimental procedure includes:

- Analysis of anomaly-centered sensor windows derived from the MetroAT anomaly detection framework developed in prior work,
- Identification of recurring anomaly patterns through clustering and temporal pattern analysis, followed by expert verification to construct subsystem-aware feature representations from the verified anomaly groups,

- Construction of subsystem-specific datasets (e.g., Brake, Air Production Unit, and Leveling datasets) based on the expert-verified subsystem-aware feature representations,
- Construction of both subsystem-aware structured representations and conventional unstructured full-feature representations,
- Implementation of selected temporal modeling approaches for failure prediction,
- Definition of consistent temporal windows and early-warning label horizons (e.g., fixed prediction horizon such as a 6-hour shifted failure labeling strategy),
- Training under temporally separated train–test splits,
- Controlled comparison between subsystem-aware structured representations and conventional full-feature representations, where all other experimental conditions (e.g., model configuration, hyperparameters, training procedure, and data splits) are kept constant in order to isolate the effect of structured subsystem-level representations,
- Quantitative evaluation using F1-score, precision, recall, and false-alarm rate, which are appropriate metrics in imbalanced and safety-critical scenarios [14].
- To enable direct comparison with the MetroAT baseline, subsystem-specific F1-scores obtained from the Brake, Air Production Unit, and Leveling datasets will be aggregated into an overall performance measure. Depending on the final distribution of samples across subsystem datasets, either a macro-average or a weighted-average F1-score will be calculated. Precision, recall, and false-alarm rate will additionally be reported for each subsystem individually to preserve subsystem-level interpretability and diagnostic insights [17].

3) Comparative and Diagnostic Analysis

To analyze the influence of subsystem-aware structured representations, models will also be evaluated on individual subsystem-based feature categories. This allows:

- Assessment of subsystem-specific predictive contribution,
- Analysis of cross-subsystem noise effects,

- Evaluation of model sensitivity to subsystem-aware versus conventional feature representations.

2. Methodical approach in research work

To structure the research workflow, this thesis follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology. CRISP-DM provides a systematic and reproducible framework suitable for applied data science research.

The research process is structured as follows:

1) **Business Understanding**

Define the predictive maintenance problem in metro systems and clarify the scope of the study, focusing on failure prediction in metro pneumatic subsystems using the MetroAT dataset.

2) **Data Understanding**

Analyze the dataset structure, investigate sensor distribution across subsystems, evaluate class imbalance, and examine temporal characteristics of sensor signals, with particular attention to anomaly patterns related to failure events.

3) **Data Preparation**

- Use anomaly-centered windows provided by the MetroAT anomaly detection framework based on temporal proximity to failure events and predefined early-warning horizons,
- Analyze anomaly-centered sensor windows,
- Apply clustering and pattern analysis techniques to identify recurring anomaly patterns,
- Verify the identified recurring anomaly groups together with domain experts from Wiener Linien,
- Construct subsystem-aware feature representations based on the expert-verified recurring anomaly groups,

- Generate subsystem-specific datasets (e.g., Brake, Air Production, and Leveling) based on the expert-verified feature representations.

These steps define the methodology for constructing subsystem-aware feature representations from recurring anomaly patterns extracted from MetroAT anomaly windows.

4) **Modeling**

- Implement selected temporal modeling approaches for time-series prediction,
- Train models under both structured (domain-grouped) and unstructured (full-feature) input configurations,
- Apply regularization and early stopping.

5) **Evaluation**

- Compare models using F1-score, precision, recall, and false-alarm rate,
- Analyze false-alarm behavior and robustness,
- Evaluate generalization performance across temporal splits,
- Aggregate subsystem-level F1-scores into an overall representation-level performance measure for comparison with the MetroAT baseline,
- Assess the influence of subsystem-aware feature representations by conducting controlled comparative experiments in which model architectures, temporal splits, training procedures, and hyperparameter settings remain constant while only the feature representation strategy is varied.

6) **Deployment-Oriented Interpretation and Analysis**

Interpret results with respect to robustness, sensitivity to input structure, and subsystem-level diagnostic contribution. Discuss how structured feature organization can support more reliable deployment in real-world predictive maintenance scenarios.

5 Relation to Completed Coursework in Data Science

The proposed thesis aims to be built directly upon methodological, computational, and analytical competencies acquired throughout the Master's program in Data Science at TU Wien. In particular, the following courses provide the academic foundation for this research:

- **Machine Learning** (184.702)
- **Advanced Methods for Regression and Classification** (105.707)
- **Experiment Design for Data Science** (188.992)
- **AKSTA Statistical Computing** (105.731)
- **Data-Intensive Computing** (194.048)
- **Information Visualization** (186.141)
- **Data-oriented Programming Paradigms** (188.995)
- **Visual Data Science** (186.868)
- **Modeling and Simulation** (194.076)
- **Statistical Simulation and Computer-Intensive Methods** (107.330)

6 Preliminary Table of Contents

- 1. Introduction
 - Motivation and Problem Statement
 - Research Objectives and Research Questions
- 2. Background and Related Work
 - Predictive Maintenance in Railway and Metro Systems
 - Temporal Learning Approaches for Multivariate Time-Series Data

- Representation Learning and Structured Feature Organization
- Robustness and False-Alarm Challenges in Real-World Predictive Maintenance
- 3. Theoretical Foundations
 - Multivariate Time-Series Analysis
 - Temporal Modeling and Sequential Learning
 - Subsystem-Aware Feature Representations
 - Failure-Related Anomaly Patterns in Predictive Maintenance
- 4. Research Design and Methodology
 - Subsystem-Aware Representation Strategy
 - Recurring Anomaly Pattern Discovery
 - Controlled Comparative Experimental Design
 - Temporal Modeling Architectures
- 5. Experimental Setup and Implementation
 - Construction of Subsystem-Aware Feature Representations
 - Baseline and Comparative Configurations
 - Training and Temporal Validation Strategy
 - Evaluation Metrics and Robustness Analysis
- 6. Results and Analysis
 - Predictive Performance Evaluation
 - Robustness and Generalization Analysis
 - Subsystem-Level Representation Analysis
 - False-Alarm Behavior Assessment

- 7. Discussion
 - Interpretation of Representation Effects
 - Practical Implications for Predictive Maintenance
 - Limitations and Future Research Directions
- 8. Conclusion
- 9. Appendix

7 Bibliography

- [1] A. Steiner, S. Helm, O. Abdelkader, and F. Ansari, “Why machine learning falls short in predictive maintenance for metro trains? real-world challenges,” in *Proceedings of the Annual Reliability and Maintainability Symposium (RAMS)*. IEEE, 2026.
- [2] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.
- [3] A. Wahid, J. G. Breslin, and M. A. Intizar, “Prediction of machine failure in industry 4.0: A hybrid cnn-lstm framework,” *Applied Sciences*, vol. 12, no. 9, p. 4221, 2022, doi: 10.3390/app12094221.
- [4] T. P. Carvalho, F. A. A. M. N. Soares, R. Vita, R. da P. Francisco, J. P. Basto, and S. G. S. Alcalá, “A systematic literature review of machine learning methods applied to predictive maintenance,” *Computers Industrial Engineering*, vol. 137, p. 106024, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360835219304838>
- [5] A. Steiner, O. Abdelkader, S. Helm, and F. Ansari, “Metroat: A benchmark dataset for data-driven maintenance in metro operations,” *Procedia CIRP*, 2025, cIRP Conference on Manufacturing Systems (CIRP CMS).
- [6] Ł. SobaŁszek, A. Gola, and A. Świć, “Time-based machine failure prediction in multi-machine manufacturing systems,” *Eksploatacja i Niezawodność*, vol. 22, no. 1, pp. 52–62, 2020.
- [7] C. Y. Park, J. W. Kim, B. Kim, and J. Lee, “Prediction for manufacturing factors in a steel plate rolling smart factory using data clustering-based machine learning,” *IEEE Access*, vol. 8, pp. 60 890–60 905, 2020, doi: 10.1109/ACCESS.2020.2983188.
- [8] F. M. Shiri, T. Perumal, N. Mustapha, and R. Mohamed, “A comprehensive overview and comparative analysis on deep learning models: Cnn, rnn, lstm, gru,” *arXiv preprint arXiv:2305.17473*, 2023.

- [9] R. Dey and F. M. Salem, “Gate-variants of gated recurrent unit (gru) neural networks,” in *2017 IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2017, pp. 1597–1600, doi: 10.1109/MWSCAS.2017.8053243.
- [10] H. Ismail Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Muller, “Deep learning for time series classification: a review,” *Data mining and knowledge discovery*, vol. 33, no. 4, pp. 917–963, 2019.
- [11] Y. He and J. Zhao, “Temporal convolutional networks for anomaly detection in time series,” in *Journal of Physics: Conference Series*, vol. 1213, no. 4. IOP Publishing, 2019, p. 042050.
- [12] Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long, “Timer: Generative pre-trained transformers are large time series models,” *arXiv preprint arXiv:2402.02368*, 2024.
- [13] M. H. Le Nguyen, “Online machine learning-based predictive maintenance for the railway industry,” Theses, Institut Polytechnique de Paris, Jun. 2023. [Online]. Available: <https://theses.hal.science/tel-04164338>
- [14] S. García-Méndez, F. de Arriba-Pérez, F. Leal, B. Veloso, B. Malheiro, and J. C. Burguillos-Rial, “An explainable machine learning framework for railway predictive maintenance using data streams from the metro operator of portugal,” *Scientific Reports*, vol. 15, no. 1, p. 27495, 2025.
- [15] C. O. Nwamekwe and O. Charles Chikwendu, “Machine learning-augmented digital twin systems for predictive maintenance in highspeed rail networks,” *International Journal of Multidisciplinary Research and Growth Evaluation*, vol. 06, no. 01, pp. 1783–1795, Feb. 2025. [Online]. Available: <https://hal.science/hal-04943345>
- [16] B. Veloso, R. Ribeiro, J. Gama, and P. Pereira, “The metropt dataset for predictive maintenance,” *Scientific Data*, vol. 9, 2022, doi: 10.1038/s41597-022-01877-3.
- [17] M. C. Hinojosa Lee, J. Braet, and J. Springael, “Performance metrics for multilabel emotion classification: comparing micro, macro, and weighted f1-scores,” *Applied Sciences*, vol. 14, no. 21, p. 9863, 2024.

8 Thesis Schedule

